Au-23-1(a)

Components of context-free grammar :

1. V is a finite set called the variables. Also called N i.e. non-terminals.

2. Σ is a finite set, disjoint from V also called T i.e. terminals.

3. R or P is a finite set of rules or productions.

4. S is the start variable, $S \in V$

Yes, there are few components that have ~~similarities~~ similarity with ~~any~~ components of regular language or finite automata.

1. Terminals are similar with the alphabet. Also both are finite set

2. Productions or rules are similar with the transition function

3. Start variable is similar in concept to the start state.

Au-23_1(a) or

a. Derivation: The sequence of substitution to obtain a string is called derivation

Parse Tree: It is a tree like structure where each node represent terminal or non terminal symbol and edge depict the sequence of substitution.

Difference between derivation and parse tree:

Derivation are represented as a linear sequence of production rules.

parse tree is a graphical structure represent the hierarchical ~~rel~~ relationship between symbols.

b. Leftmost derivation: Applying production to the leftmost variable in each step

Righmost derivation: Applying production to the rightmost variable ~~a~~ in each step.

Difference between leftmost and rightmost derivation

Both derivation result in the same string but they differ in the order in which non terminals are expanded during the derivation process.

Au-23_1(b)

Ambiguity: In context free grammar ambiguity refers to a situation where a grammar generates the same string in several different ways.

Given grammar:

$S \longrightarrow AB$

$A \longrightarrow aA \mid abA \mid \varepsilon$

$B \longrightarrow bB \mid abB \mid \varepsilon$

$S \rightarrow AB$	$S \rightarrow AB$
$\rightarrow abAB$ [$A \rightarrow abA$]	$\rightarrow aAB$ [$A \rightarrow aA$]
$\rightarrow abAabB$ [$B \rightarrow abB$]	$\rightarrow aB$ [$A \rightarrow \varepsilon$]
$\rightarrow ababB$ [$A \rightarrow \varepsilon$]	$\rightarrow abB$ [$B \rightarrow bB$]
$\rightarrow abab$ [$B \rightarrow \varepsilon$]	$\rightarrow ababB$ [$B \rightarrow abB$]
	$\rightarrow abab$ [$B \rightarrow \varepsilon$]

The given grammar is ambiguous because the string "abab" derived ambiguously in that grammar.

Au-23_1(c) & 1(c) or

Given grammar,

$S \longrightarrow SSx \mid SSy \mid SSz \mid a \mid b \mid c$

Left most derivation:

$S \rightarrow SSz \quad [S \rightarrow SSz]$

$\rightarrow SSySz \quad [S \rightarrow SSy]$

$\rightarrow cSySz \quad [S \rightarrow c]$

$\rightarrow cSSxySz \quad [S \rightarrow SSx]$

$\rightarrow caSxySz \quad [S \rightarrow a]$

$\rightarrow cabxySz \quad [S \rightarrow b]$

$\rightarrow cabxycz \quad [S \rightarrow c]$

Rightmost derivation:

$S \rightarrow SSz \quad [S \rightarrow SSz]$

$\rightarrow Scz \quad [S \rightarrow c]$

$\rightarrow SSycz \quad [S \rightarrow SSy]$

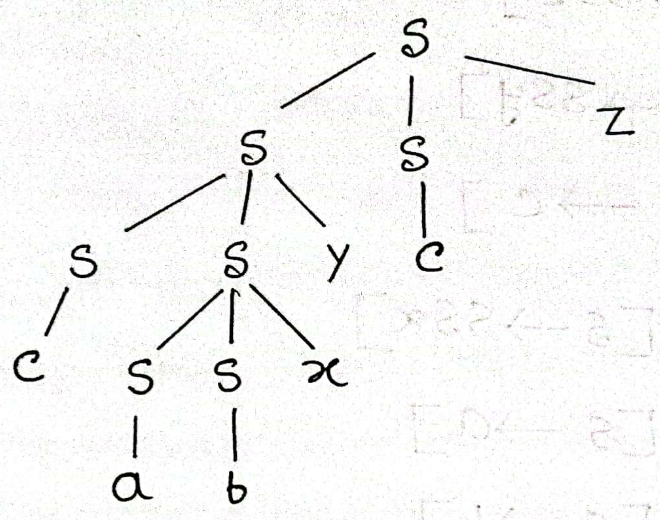
$\rightarrow SSSxycz \quad [S \rightarrow SSx]$

$\rightarrow SSbxycz \quad [S \rightarrow b]$

$\rightarrow Sabxycz \quad [S \rightarrow a]$

$\rightarrow cabxycz \quad [S \rightarrow c]$

Parse tree for the string:



Au-23 - ~~2(b)~~ 2(a) or

a) $0^* 1^*$

CFG:

$$S \rightarrow 0S1 \mid \epsilon$$

b) $1(01)^*$

CFG:

$$S \rightarrow 1A$$

$$A \rightarrow 01A \mid \epsilon$$

c) $(11 \cup 0)^*$

CFG:

$$S \rightarrow 11S \mid 0S \mid \epsilon$$

Au-23_2(b)

i) $L = \{a^{2n} b^m c^n \mid n \geq 0\}$

$A \longrightarrow aaAc \mid B \mid \varepsilon$

$B \longrightarrow bB \mid \varepsilon$

ii) $L = \{a^m b^n c^{m+n} \mid n \geq 0\}$

$A \longrightarrow aAc \mid B \mid \varepsilon$

$B \longrightarrow bBc \mid \varepsilon$

Au-23_2(c)

$R \longrightarrow aSa \mid bRb \mid S$

$S \longrightarrow aTb \mid bTa \mid aS$

$T \longrightarrow XTX \mid X \mid \varepsilon$

$X \longrightarrow a \mid b$

Step-1: Make a new start variable

$R_0 \to R$

$R \to aSa \mid bRb \mid S$

$S \to aTb \mid bTa \mid aS$

$T \to XTX \mid X \mid \varepsilon$

$X \to a \mid b$

Step-2: Remove null production

$R_0 \to R$

$R \to aSa \mid bRb \mid S$

$S \to aTb \mid bTa \mid aS \mid ab \mid ba$

$T \to XTX \mid X \mid XX$

$X \to a \mid b$

Step-3: Remove unit rules

$R_0 \to aSa \mid bRb \mid aTb \mid bTa \mid aS \mid ab \mid ba$

$R \to aSa \mid bRb \mid aTb \mid bTa \mid aS \mid ab \mid ba$

$S \to aTb \mid bTa \mid aS \mid ab \mid ba$

$T \to XTX \mid a \mid b \mid XX$

$X \to a \mid b$

Step-4|

$R_0 \rightarrow XSX \mid XRX \mid XTX \mid XTX \mid XS \mid XX \mid XX$

$R \rightarrow XSX \mid XRX \mid XTX \mid XTX \mid XS \mid XX \mid XX$

$S \rightarrow XTX \mid XTX \mid XS \mid XX \mid XX$

$T \rightarrow XTX \mid a \mid b \mid XX$

$X \rightarrow a \mid b$

Step-5|

$R_0 \rightarrow XSX \mid XRX \mid XTX \mid XS \mid XX$

$R \rightarrow XSX \mid XRX \mid XTX \mid XS \mid XX$

$S \rightarrow XTX \mid XS \mid XX$

$T \rightarrow XTX \mid a \mid b \mid XX$

$X \rightarrow a \mid b$

Step-6|

$R_0 \rightarrow XU_1 \mid XU_2 \mid XU_3 \mid XS \mid XX$

$R \rightarrow XU_1 \mid XU_2 \mid XU_3 \mid XS \mid XX$

$S \rightarrow XU_3 \mid XS \mid XX$

$T \rightarrow XU_3 \mid a \mid b \mid XX$

$U_1 \rightarrow SX$

$U_2 \rightarrow RX$

$U_3 \rightarrow TX$

$X \rightarrow a \mid b$

This is the required chomsky normal form

Au-23_3(a)

We remove an ε rule $A \rightarrow \varepsilon$ where A is not the start variable. Then for each occurance of an A on the right-hand side of a rule, we add a new rules with that occurance deleted. If $R \rightarrow uAv$ then we add rule $R \rightarrow uv$ so for rule $R \rightarrow uAvAw$ we add $R \rightarrow uvAw$ and $R \rightarrow uAvw$ and $R \rightarrow uvw$. If the rule $R \rightarrow A$ then we add $R \rightarrow \varepsilon$

Au-23_3(a) or

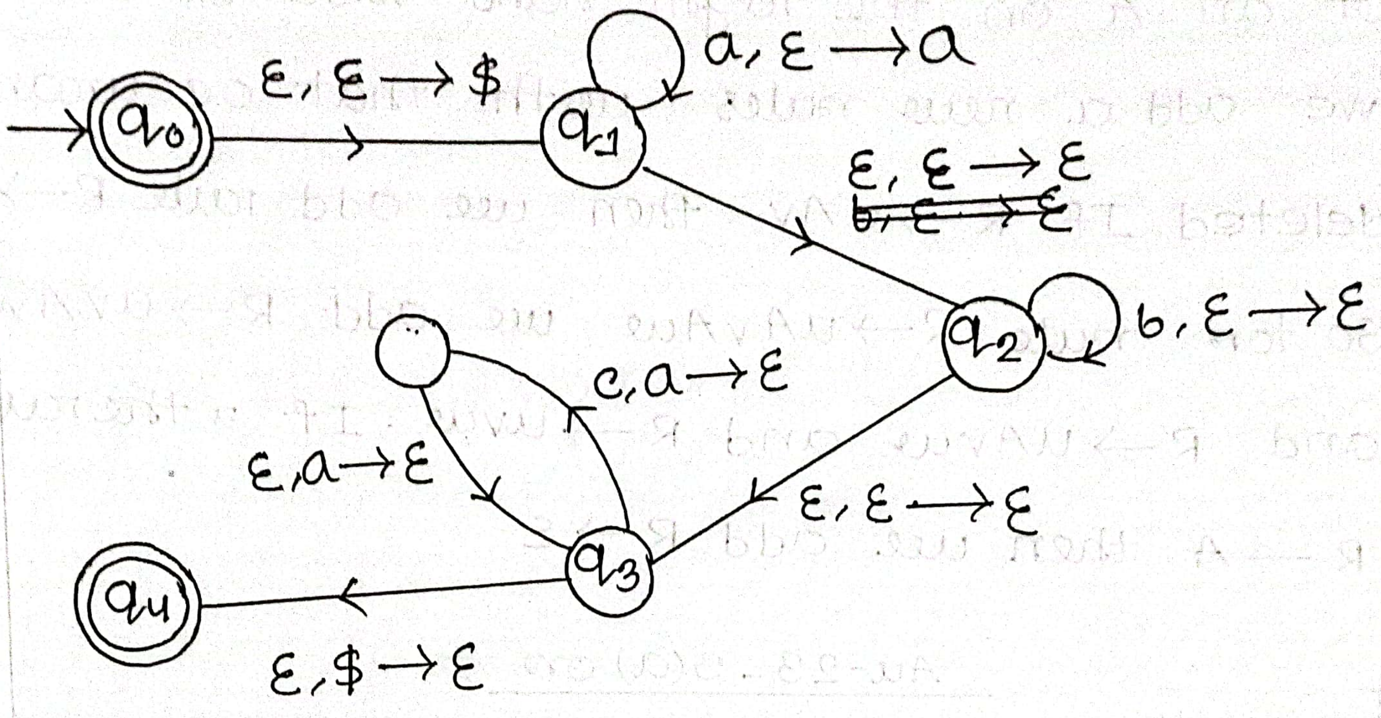
Pushdown automata are more powerful than finite automata because it have an extra component called a stack. The stack provide additional memory beyond the finite amount available in control. The stack allows pushdown automata to recognize some non regular languages.

CamScanner

Au-23_3(b)

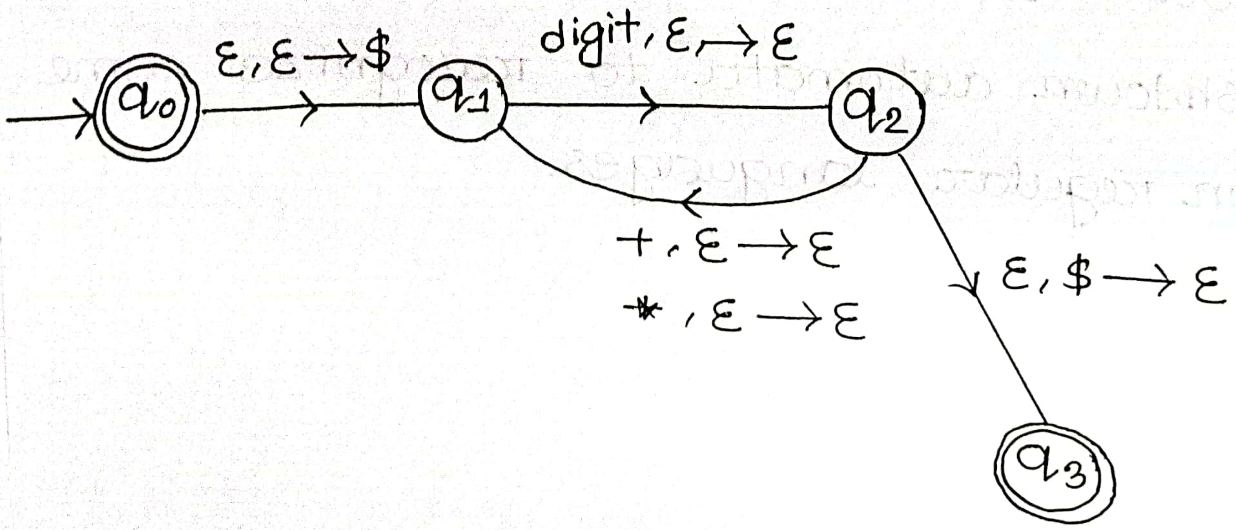
Given language : $L = \{a^{2n} b^m c^n \mid n \geq 0\}$

Pushdown automata :



Au-23_3(b) or

A pushdown automata that recognize any arithmatic expression involving +, * and any one digit integer



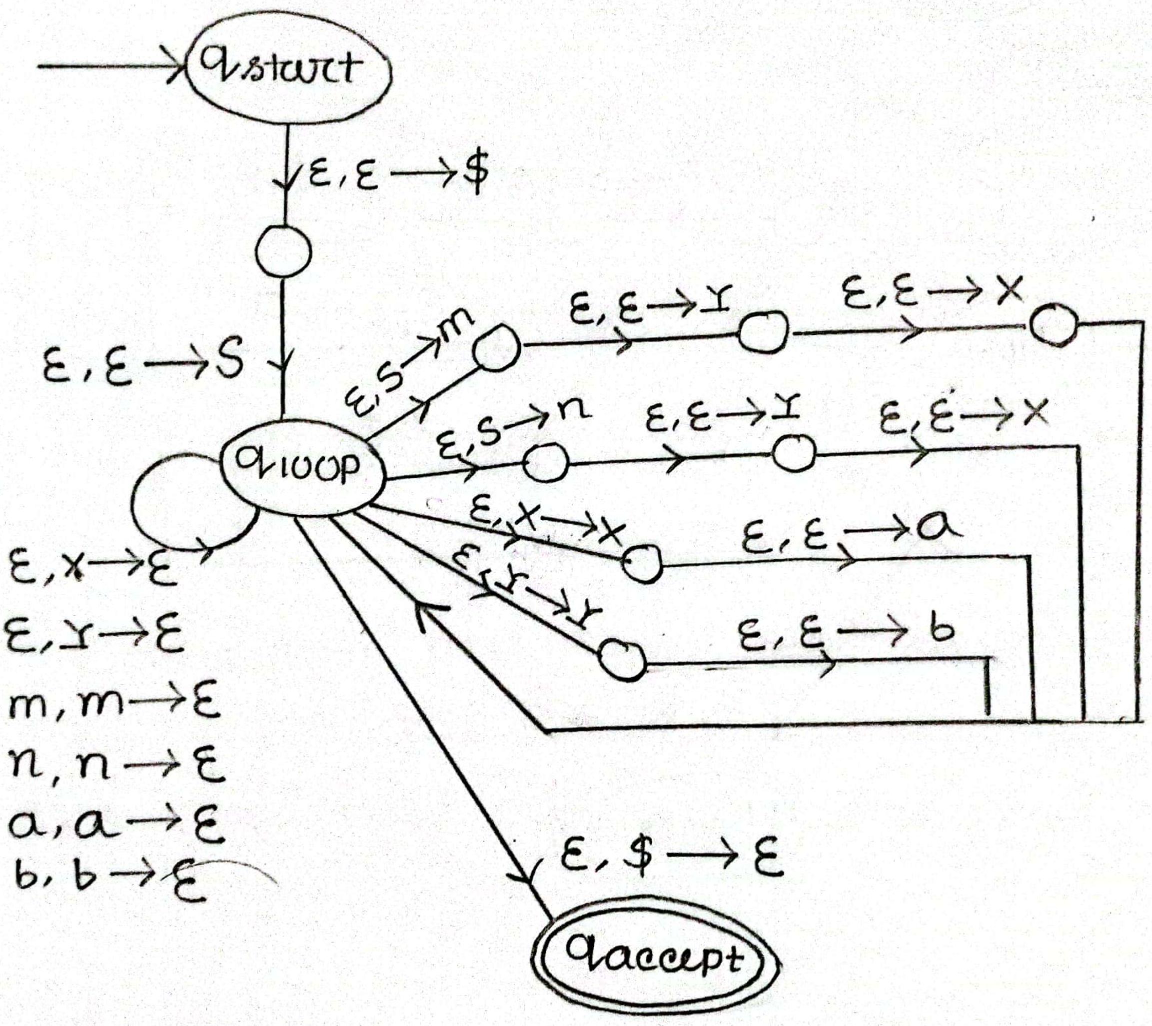
Au-23_3(C)

CFG to PDA

$S \rightarrow XYm \mid XYn$

$X \rightarrow aX \mid \varepsilon$

$Y \rightarrow bY \mid \varepsilon$



Au-23- 3(C) OR

$S \longrightarrow aXbY$

$X \longrightarrow aYa \mid \varepsilon$

$Y \longrightarrow bXb \mid c \mid \varepsilon$